

DataFest 2025: Analyzing Microtrends within Savills' Real Estate Data

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Abstract

This research leveraged a commercial real estate dataset provided by Savills to analyze key market trends and identify actionable insights for their advisory services. The analysis integrated lease data, market occupancy data, and unemployment statistics to explore factors influencing space availability, revenue, and rent trends. A significant finding was the correlation between unemployment rates and available commercial space, where regions with lower unemployment showed reduced space availability, signaling increased demand. Linear regression models forecasted substantial rent growth for specific industries, including Coworking and Executive Suite companies, from 2024 to 2026. The results could provide Savills with data-driven recommendations for focusing on high-demand regions, such as Manhattan, and industries poised for rent increases, potentially assisting with more informed leasing decisions for their clients.

Keywords: Commercial Real Estate, Leasing Activity, Rent Trends, Space Availability, Unemployment

1 Introduction

Savills, a prominent international commercial real estate firm, plays a crucial role in assisting businesses with their office space requirements. The firm's advisory services are built on a variety of data sources that provide information on market trends, regional growth, and availability of office space. Given the lack of publicly accessible websites for commercial real estate data, Savills relies on the compilation and analyzing of data from multiple sources to inform their recommendations. This research leverages a dataset provided by Savills to identify notable trends and microtrends in the commercial real estate market. The goal is to provide actionable insights that Savills can use to guide clients in making informed decisions about where, when, and how to lease office space.

The data sets used in this competition consisted of several different files.

1. Leases: This dataset, sourced directly from Savills and compiled from multiple providers, including CoStar and Savills, contains records of lease transactions. Each row represents a business signing a lease agreement with a landlord, granting them the right to occupy a property. The dataset includes all recorded lease transactions from January 1, 2018, to December 31, 2024. In addition to individual leases, the dataset contains aggregated market-level information, such as the total available rental space within a market for a given quarter. These variables are often segmented based on internal building classifications: Class A (internal class = A) - High-quality buildings and Class O (internal class = O) - Other buildings that do not meet Class A standards. For example, the RBA (Rentable Building Area) variable represents the total available rental space in a specific market and quarter for each internal class. Within a market (e.g., Atlanta), the same value appears for all lease transactions of Class A buildings in a given quarter, and a separate value exists for Class O buildings.
2. Major Market Occupancy Data: This dataset, sourced from Kastle, a security company specializing in commercial building security, provides estimated building occupancy rates based on card-swipe activity from 2020 to 2024. Only 10 markets are included. The ten markets are Austin, Chicago, Dallas/Ft.Worth, Houston, Los Angeles, Manhattan, Philadelphia, San Francisco, South Bay/San Jose, and Washington D.C. Savills acquired this data to track occupancy trends. Some key considerations to consider: Occupancy rates are reported as percentages relative to a baseline set on March 1, 2020. A 100%' occupancy rate means the building has the same occupancy as it did on March 1, 2020, not that all units are occupied.
3. Unemployment.csv: This dataset was taken directly from the U.S. Bureau of Labor Statistics. Each row provides data on the percent unemployment within a state for a given year (2014-2024).

2 Methods

We started by pre-processing and combining the data sets to allow for more consistency in our analysis. One of the challenges of data analysis is that our data sets contain many missing values. To address this, for numerical columns, missing values were filled using the median, as it is less sensitive to outliers. For categorical columns, missing values were filled using the mode, representing the most frequently occurring value. After preprocessing, we merged the data sets to create a comprehensive view of the market. Lease data was merged with Market Data based on year, quarter, and market, ensuring occupancy proportions were aligned with lease transactions.

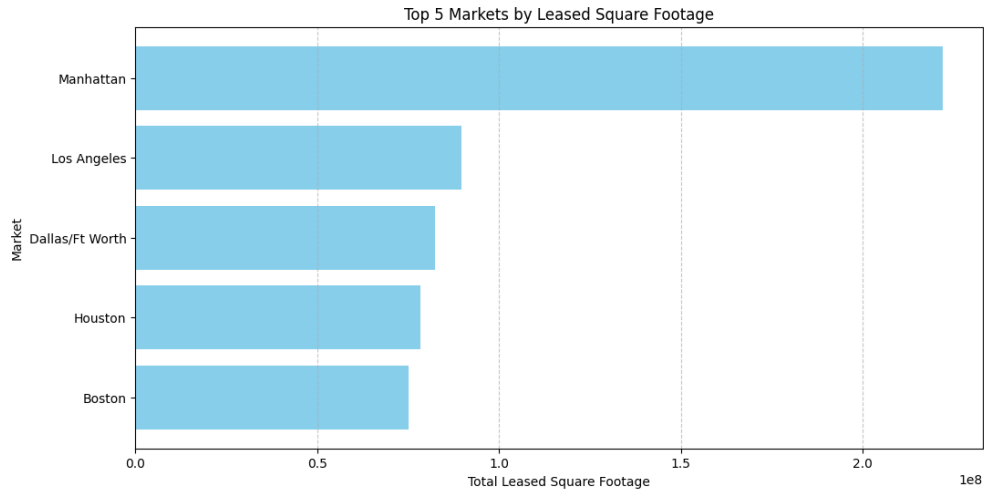


Fig. 1 Top 5 Markets by Leased Square Footage

We wanted to examine particular regions and industries within our dataset that could be used for further analysis. The top market for leased square footage was Manhattan, followed by Los Angeles, Dallas, and Fort. Worth, Houston, and Boston.

The top industry by leased square footage was Financial Services and Insurance, followed by Technology/Advertising, Business/Consulting, Legal Services and Government.

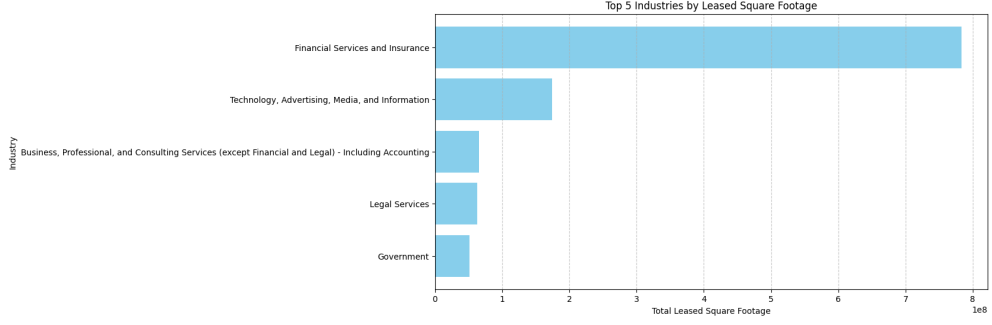


Fig. 2 Top 5 Industries by Leased Square Footage

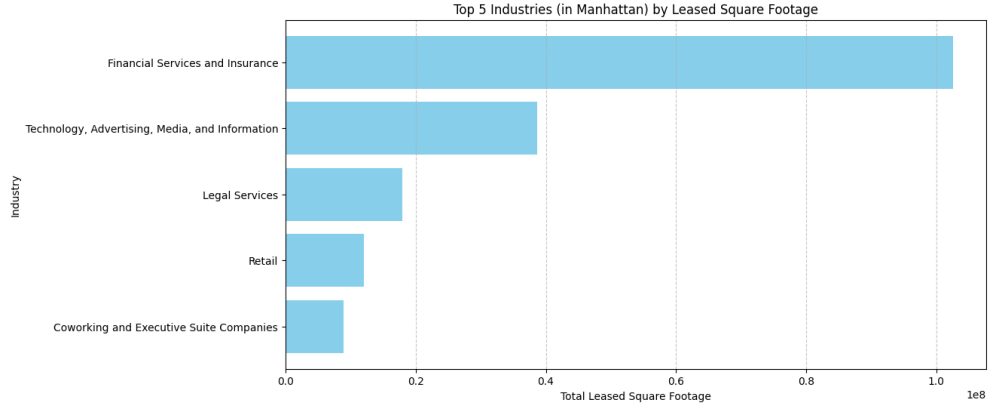


Fig. 3 Top 5 Industries in Manhattan by Leased Square Footage

We looked further into the Manhattan market specifically to find particular industries of interest. As seen in Figure 3, the top industry in Manhattan by leased square footage was Financial Services and Insurance, followed by Technology/Advertising, Legal Services, Retail, and Coworking/Executive Suite Companies.

To assess the financial impact of leasing activity, we estimated revenue for each lease transaction. The revenue calculation was based on leased square footage (leasedSF) and rental rates, adjusted according to building classification.

$$\text{Revenue} = \text{leasedSF} \times \begin{cases} \text{internal_class_rent}, & \text{if internal_class} = 'A' \\ \text{overall_rent}, & \text{otherwise} \end{cases} \quad (1)$$

The top market based on revenue was Manhattan, followed by Los Angeles, Boston, San Francisco and Dallas/Ft. Worth.

The unemployment.csv dataset was organized by state and not market, so we needed to find a way to append unemployment data to compare between other categories. To do so, we created a state column in our combined dataset by developing

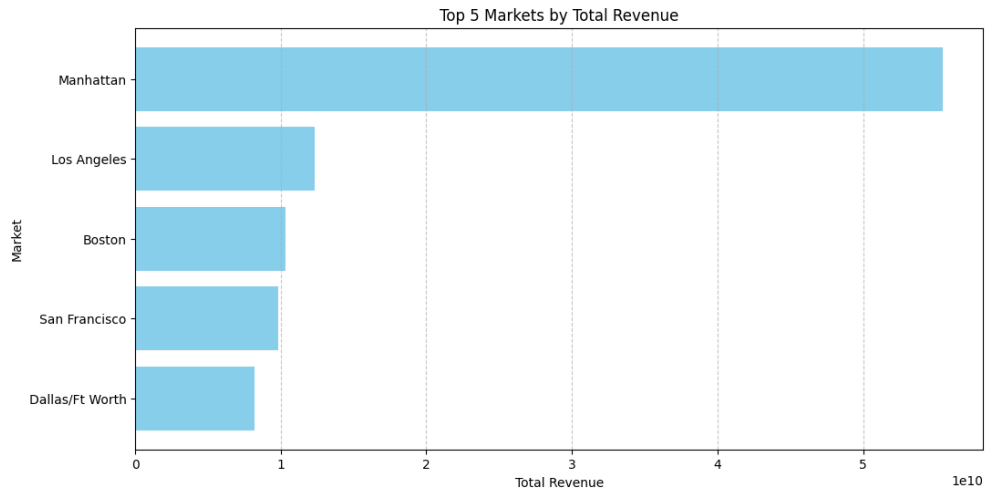


Fig. 4 Top 5 Markets by Total Revenue

a dictionary to map markets based on their respective states. We then merged the unemployment data based on the matching state, month, quarter and year. Table 1 shows the dictionary mapping between the respective markets and their states.

3 Results

3.1 Linear Regression Analysis between Categorical Variables

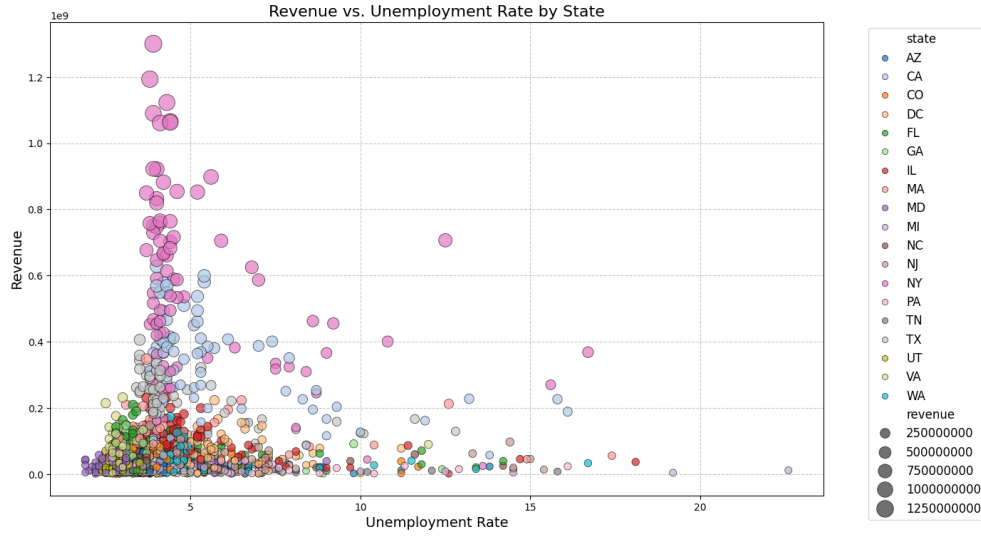


Fig. 5 Scatterplot of Revenue vs. Unemployment Rate

One of our first goals was determining if there was a relationship between revenue and unemployment rate, as we wanted to see if unemployment rates in each state were having an effect on total company earnings. Figure 5 shows that no specific relationship exists between the two variables. However, the majority of data from New York state is concentrated with high revenue and low unemployment, suggesting that the region could be benefiting from a robust economy with stable employment levels, potentially leading to higher demand for commercial real estate and, consequently, higher revenues for Savills. This may also indicate that New York's economic conditions support more leasing activity, which in turn could contribute to Savills' increased revenue. We further analyzed the state of New York to investigate whether the unemployment rate was influencing other factors, such as space availability. Figure 6 shows that while some regions in New York show a pattern of low unemployment and low availability of space, a general negative correlation was observed between the unemployment rate and space availability. This suggests that as unemployment decreases, available commercial real estate space tends to become more limited, potentially indicating higher leasing demand and occupancy rates.

In contrast, as seen in Figure 7, when examining factors contributing to space availability, we found that overall rent was positively associated with available space. This suggests that in markets with higher rent prices, there is more space available,

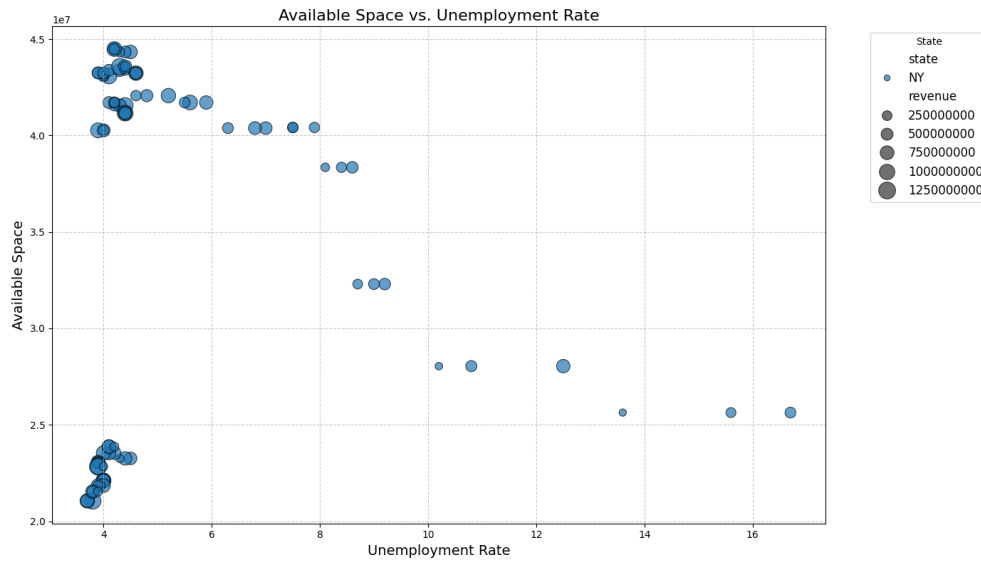


Fig. 6 Available Space vs. Unemployment

potentially due to tenants seeking more affordable or spacious options, or landlords providing additional space to meet demand.

Similarly, Figure 8 shows that leasing activity exhibited a positive association with space availability, which aligns with the expectation that higher leasing activity corresponds with increased demand for commercial space. These findings suggest that while unemployment may negatively impact space availability, other factors, such as rent levels and leasing activity, are also significant contributors to the overall supply of space.

Additionally, we conducted an Ordinary Least Squares (OLS) multivariate regression to understand the factors influencing available space in commercial real estate. The regression results, as seen in Figure 9, provide several key insights. The R squared value of 0.447 indicates that approximately 44.7% of the variance in available space is explained by the model. Additionally, the f-statistic was 12.62 with a p-value of $5.45e-09$ is significant, suggesting that at least one of the predictors in the model is statistically significant in explaining the variation in available space. The variables of unemployment_rate, overall_rent, and leasing show significant association with available_space with p-values of less than 0.05. However, a discrepancy exists between the coefficients and the trend seen in Figures 7 and 8, suggesting that the variables could be confounding the results.

3.2 Prediction Forecasting of Key Industries

Finally, we wanted to see how industries were shifting in rent over time, and forecast rent trends based on their average rent prices. The primary objective was to identify trends in rent prices for these industries and project future rent values from 2024

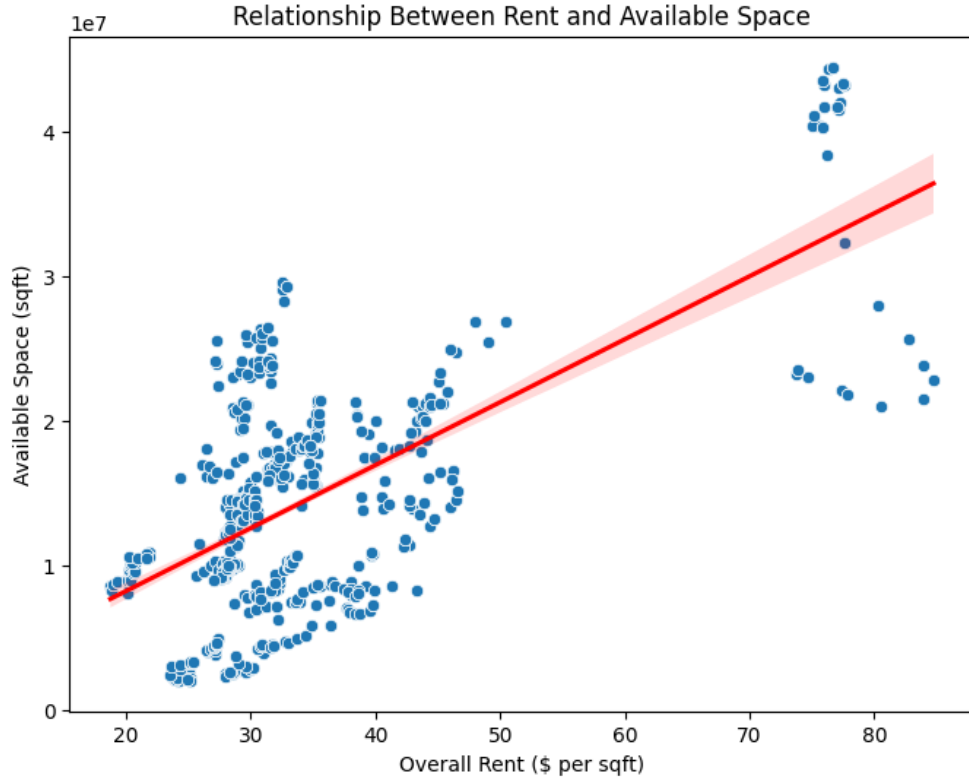


Fig. 7 Rent vs. Available Space

to 2026 using linear regression. To begin, we computed the average rent for each industry by grouping the dataset by `internal_industry` and calculating the mean of the `internal_class_rent` variable. From this, we identified the top five industries with the highest average rent prices, which were selected for further analysis. For each of the top five industries, we employed linear regression to model the relationship between rent prices and years. Linear regression was chosen for its simplicity and effectiveness in forecasting time-based data. The year was used as the feature (independent variable), and the `internal_class_rent` was the target (dependent variable). We trained the linear regression model using the historical data (years and corresponding rent prices) for each industry. Following the model training, we forecasted future rent prices for the years 2024 through 2026, providing insights into how rent prices might evolve based on historical trends.

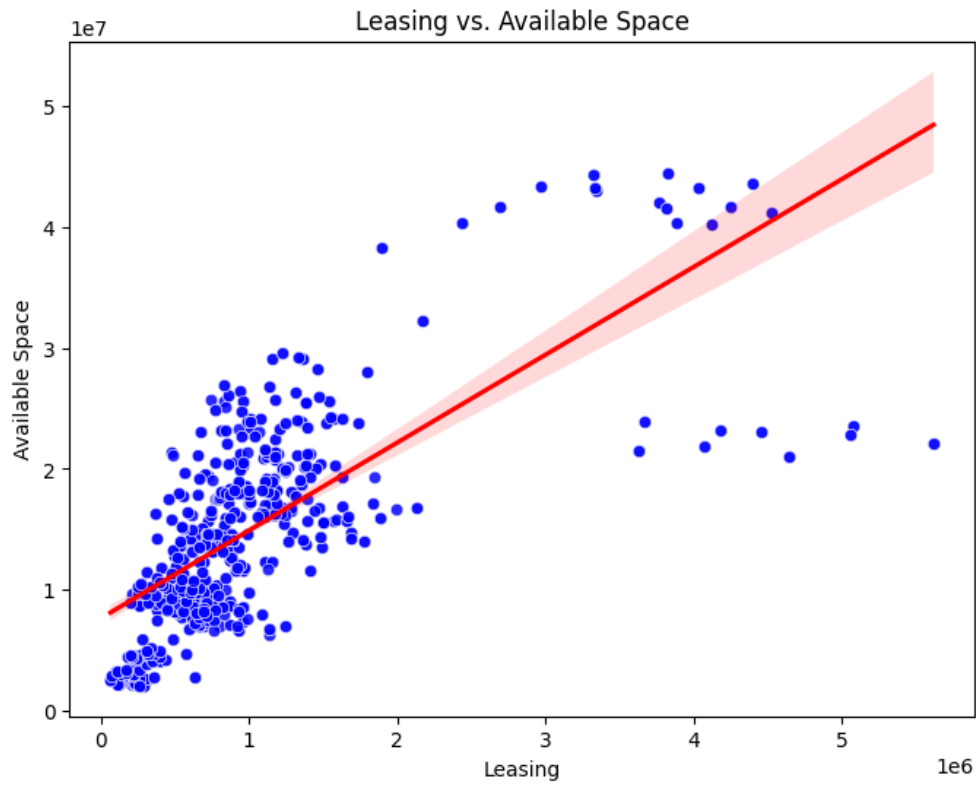


Fig. 8 Leasing vs. Available Space

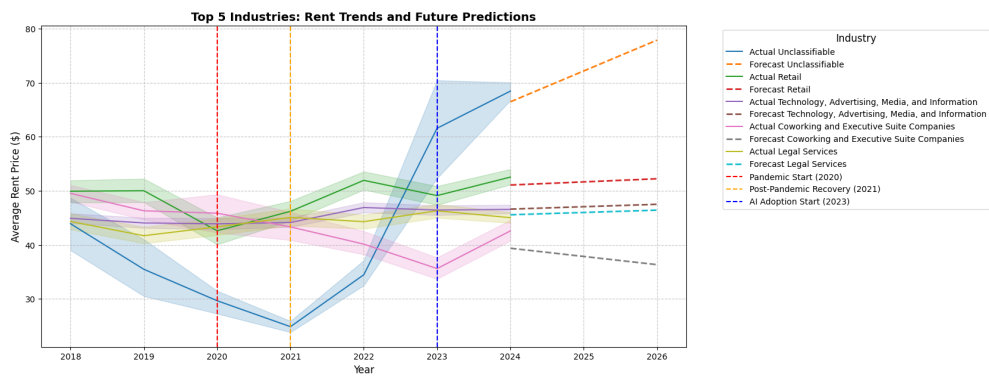


Fig. 10 Prediction of Top 5 Industries of Average Rent Price

OLS Regression Results

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Dep. Variable:

available_space

R-squared:

0.447

Model:

OLS

Adj. R-squared:

0.412

Method:

Least Squares

F-statistic:

12.62

Date:

Sun, 23 Mar 2025

Prob (F-statistic):

5.45e-09

Time:

04:51:17

Log-Likelihood:

-1441.2

No. Observations:

84

AIC:

2894.

Df Residuals:

78

BIC:

2909.

Df Model:

5

Covariance Type:

nonrobust

=====

coef

std err

t

P>|t|

[0.025

0.975]

=====

const

1.706e+08

2.36e+07

7.234

0.000

1.24e+08

2.18e+08

unemployment_rate

-1.318e+06

5.14e+05

-2.563

0.012

-2.34e+06

-2.94e+05

revenue

0.0052

0.003

1.570

0.120

-0.001

0.012

overall_rent

-1.333e+06

3.03e+05

-4.402

0.000

-1.94e+06

-7.3e+05

leasing

-5.8292

1.490

-3.912

0.000

-8.796

-2.863

leasedSF

-550.3034

434.468

-1.267

0.209

-1415.263

314.656

=====

Omnibus:

6.351

Durbin-Watson:

0.255

Prob(Omnibus):

0.042

Jarque-Bera (JB):

5.575

Skew:

-0.547

Prob(JB):

0.0616

Kurtosis:

2.370

Cond. No.

1.91e+10

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Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.91e+10. This might indicate that there are strong multicollinearity or other numerical problems.

Fig. 9 Regression Results

As seen in Figure 10, our forecasting analysis showed a substantial predicted growth for Unclassifiable industries, with it surpassing all other industries for the highest average rent price.

To better understand the temporal dynamics of rent changes, we computed a new variable called `rent_pct_change`; the percentage change in rent for each industry. This was done by calculating the percentage change between consecutive years. Following the computation of rent percentage changes, we filtered the dataset to retain only the top five industries, based on the highest average rent prices. This subset of the data was used for the linear regression analysis. For each of the top five industries, we trained a linear regression model to predict rent percentage changes based on the year. The year was used as the independent variable (feature), while the rent percentage change served as the dependent variable (target). After fitting the model, we used it to predict the future rent percentage changes for the years 2024 through 2026.

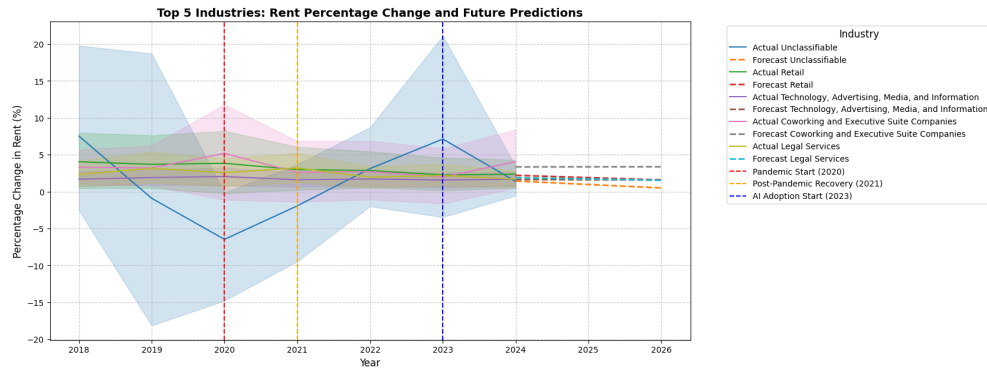


Fig. 11 Prediction of Top 5 Industries of Percentage Change in Rent

One noticeable trend we saw, as shown in Figure 11, was a fluctuation in the percentage change in rent of Unclassifiable industries, as they shifted up and down from 2018-2020 and 2020-2023 respectively, with a decrease in 2024. Our model forecasted an increase in Coworking and Executive Suite Companies with the highest percentage change compared to the other industries.

4 Conclusion

The analysis of the commercial real estate market data yielded several key insights and actionable findings that could inform Savills' advisory services. For one, New York seemed to be a major region of interest, as Manhattan exhibited the highest leased square footage and total revenue amongst all markets. New York state also showed a trend of high revenue and low unemployment, indicating a strong economy that may support higher leasing activity and demand for commercial space.

Additionally, a negative correlation was observed between unemployment rates and space availability: as unemployment decreased, available commercial real estate space became more limited. This suggests higher leasing demand in regions with low unemployment. Rent levels and leasing activity were positively correlated with space availability, indicating that higher rents and increased leasing activity are often associated with higher demand for commercial space.

Finally, linear regression analysis of rent trends revealed substantial predicted growth for "Unclassifiable" industries, which had the highest average rent prices over time. Coworking and Executive Suite companies were expected to experience the highest percentage increase in rent from 2024 to 2026. Forecasts suggest that rent prices in these industries will continue to rise, possibly driven by shifting market preferences or changing demand dynamics.

To leverage these findings and provide more guidance to clients while maximizing profit margins, Savills should look to promote more real estate in areas with strong leasing demand and low space availability, such as New York. Additionally, they should consider looking into industries that may experience rent increases in the future, such as Coworking and Executive Suite companies. Understanding the economic factors influencing commercial real estate markets (e.g., unemployment rates, rent levels) can allow Savills to tailor their advice to clients based on current and projected market conditions.